# Reward-Driven Decision Making

## USING REINFORCEMENT LEARNING TO OPTIMIZE ROUTES

Richard Hoehn | March 7th, 2026

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Reward-Driven Decision Making: Using Reinforcement Learning (RL) to Optimize Routes

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# ABOUT ME & THE PROBLEM

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## Practical Experience & Applied Science Research

I work at **Aerton, Inc.** focused on Logistics & Field Service software for rural team.

We manage internal Fleet routing, dispatching, and tracking.

I'm also a PhD Student at MTSU - Computational Data Sciences.

My research is on Reinforcement Learning (RL) & Optimization models.

My research is Applied Sciences – Meaning finding solutions of current real-world problems.

The real-world environment is messy!

Our customers are trying to find repeatable operational ways to be better routing decisions.

In short, they strive to find an optimal solution to ever changing requirements.

Most businesses face these problems.

The difficulty is that the problems are a sequence of decisions not a single decision.

**Solving the Problem is Sequential**

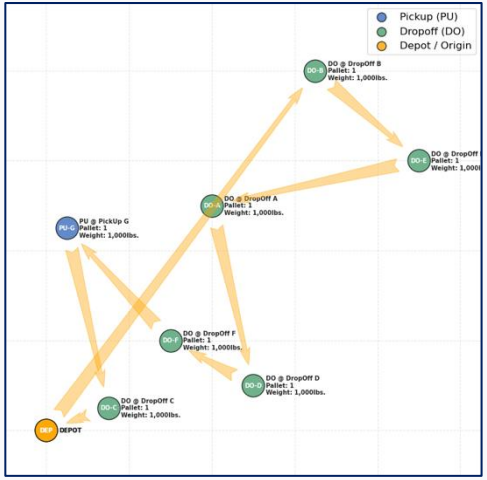
# OPTIMIZATION IS EVERYWHERE

## Why “Shortest Path” Often Fails

Real operations involve trade-offs, not just <u>distance!</u>

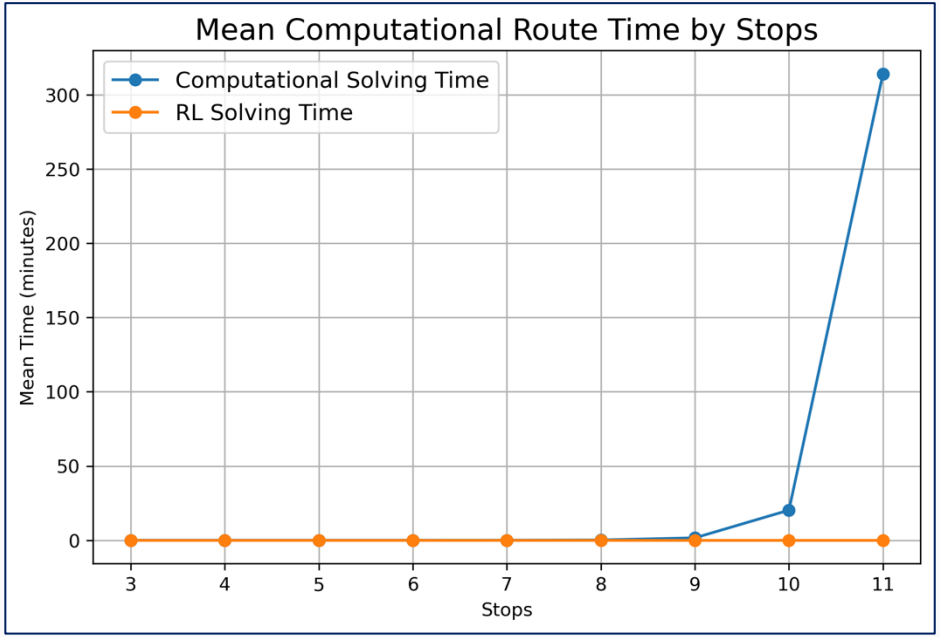
The closest next step is not always the best overall step.

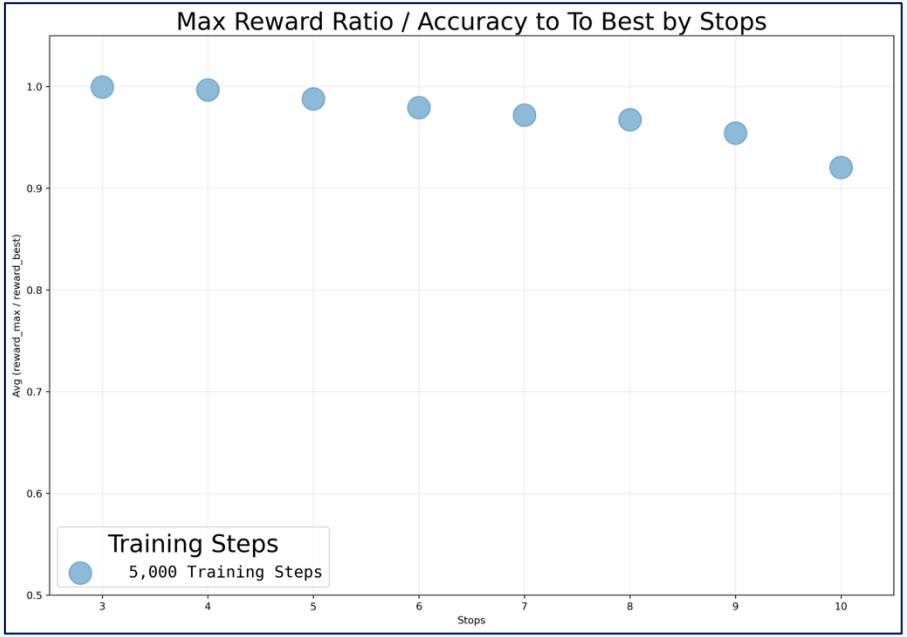
Constraints and <u>downstream</u> consequences matter!



# WHY RL RESEARCH – COMPUTATIONAL COST

## The Computational Cost of the Best Might be In-Feasible for Large Routes



# WHAT IS REINFORCEMENT LEARNING?

## Learning Decisions from Experience

RL is a Subset of Machine Learning ML.

A model that allows a system to learn decisions from experience.

RL tries Actions and receives feedback in form of rewards.

It gradually improves its decision making over time.

It learns by interacting, not just being told the answer.

**BOTTOM LINE**

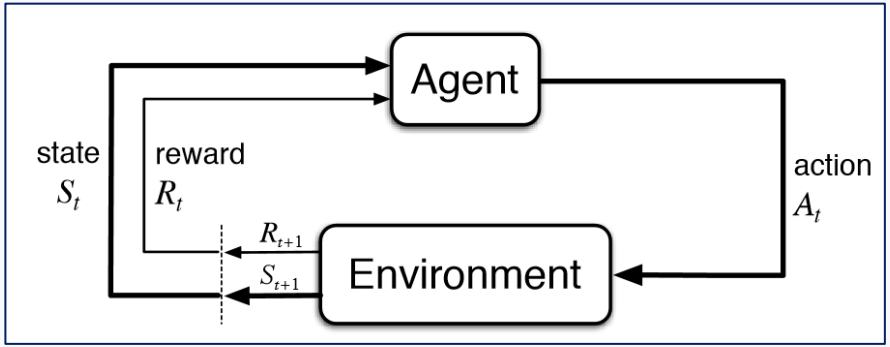
**Learn by interacting, not just being told the answer.**

**Receive reward for good behavior!**

# THE CORE RL LOOP — WHAT IS THE AGENT

## A Continuous Cycle of Improvement

1. **Observe** the current state
2. Choose an **Action**
3. Receive Feedback / **Reward**
4. Update behavior - Learn
5. Repeat




OBSERVE
↓
CHOOSE ACTION
↓
RECEIVE REWARD
↓
UPDATE
↓
REPEAT

# RL QUESTIONS THAT MATTER

## Defining System Boundaries in Reinforcement Learning

**What can the system see?**

Observation Space

**What can the system do?**

Action Space

**How do we measure?**

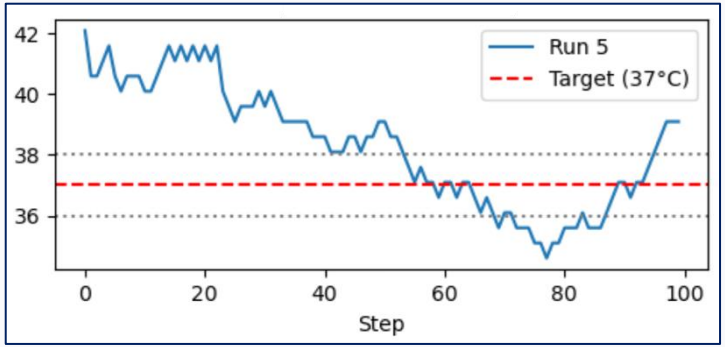
Reward

# INTUITION: SIMPLE THERMOSTAT EXAMPLE

## A Simple Example to Build Intuition on RL

### PRIMARY GOAL

Keep the room near a target temperature while avoiding energy waste.




### OBSERVATION

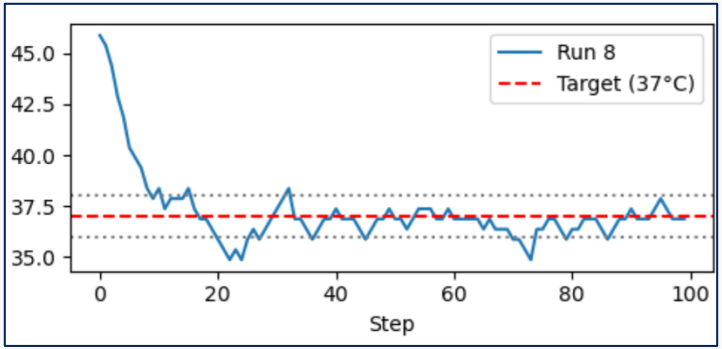
Current, target, and outside temperatures

### ACTION

Increase | Decrease | Maintain Heat

### REWARD

+ For target Proximity



# SCALING COMPLEXITY – REAL WORLD APPS

## What Changes When the Problem Gets Harder?

- More Information to Process – What is the State of Everything?
- More Possible Actions
- Stronger Long-Term Effects
- Increasing Number of Operational Constraints

### SYSTEM SCALING

**Better decisions become less obvious as the state space grows.**

***Especially for Humans!***

# ROUTE OPTIMIZATION IN AG IS HARD

## Real-World Constraints in Rural Logistics

### Pick-Up & Drop-Off Limitations!

We only have 20-26 Pallet Spots → Drop-Off First!

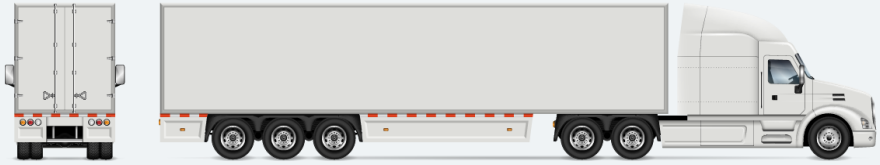
### DOT Hours of Operation Constraints

Only allowed to drive 11hours

### Fuel Costs Under Heavy Loads (*See Example*)

With heavy loads the fuel consumption goes up

| Length | Width | Height | Volume | Max Cargo Weight | Pallets |
|---|---|---|---|---|---|
| 48–53 ft | Up to 102.36 in | 13–14 ft | Up to 3,489 cu ft | 45,000 lbs | 26 |



**OPERATIONAL IMPACT**

**Cost, time, and service all matter simultaneously**



Image Source: https://www.directliquidation.com

# SEQUENCE QUALITY vs. DISTANCE / TIME

## The best Route is Not Always the Shortes

A closer stop may create a **worse route later**!

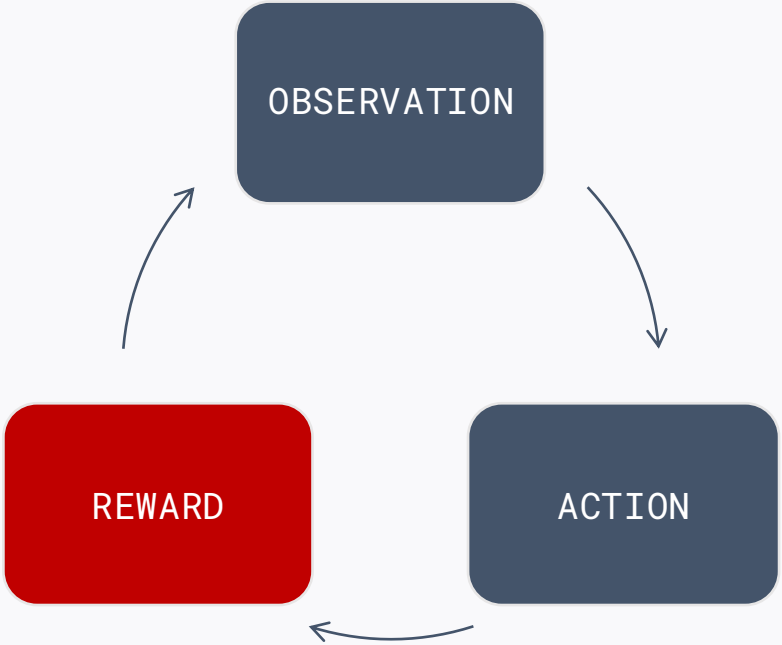
Pick-Up(s) increase Load / Weight
Drop-Off(s) reduce Load / Weight
→ **Sequence matters!**

Fuel and time **costs** change based on **sequence**.

# MAPPING LOGISTICS TO REAL-WORLD

## How Real-World Route Planning Becomes an RL Problem

### The Secret is...



# FEEDBACK

Reward-Driven Decision Making: Using Reinforcement Learning (RL) to Optimize Routes

# REWARD SHOULD MATCH REAL GOAL

## Aligning Feedback with Business Value!

**Give Reward on Successful Drop-Off & Pick-Up:**
This can be as simple as the count of Pallets / Weight or the actual revenue per load.

**Penalize Time & Distance Costs:**
This can is based on hourly rate of driver. Cost of Truck per Mile.

**Penalize Fuel Consumption based on Weight:**
Since we have the state of the truck we can estimate the fuel consumption based on the weight being hauled.

$$r_t = \underbrace{P(a_t)}_{\text{DO/PU Profit}} - \underbrace{C_{\text{drive}}\,\Delta time_t}_{\text{Drive Time Cost}} - \underbrace{C_{\text{gallon}}\,\frac{\Delta distance_t}{\text{MPG}(W_t)}}_{\text{Fuel Cost}}$$

**BUSINESS ALIGNMENT**

The reward function is the **Bridge** between business objectives and an RL-Model behavior

# HOW IT LEARNS

## The Iterative RL Training Process

$$r_t = \underbrace{P(a_t)}_{\text{DO/PU Profit}} - \underbrace{C_{\text{drive}}\,\Delta time_t}_{\text{Drive Time Cost}} - \underbrace{C_{\text{gallon}}\,\frac{\Delta distance_t}{\text{MPG}(W_t)}}_{\text{Fuel Cost}}$$

**RL is un-supervised learning process!**

**Generate Route Scenarios – Random / Real**
Let the model run an episode in a simulated environment

**Measure the Resulting Rewards**
The reward are used for internal policy updates and back propagation

**Repeat across Many Scenarios and Episodes**
In training we ran over 500 scenarios with 5,000 episodes each

TRAINING CYCLE

**Learning is a continuous cycle of trial, feedback, and adjustment.**

# REWAWRD | Action | OBSERVATION

## Python Code

```python
# ----------------------------
# 4) Completion bonus (only on first depot return after all serviced)
# ----------------------------
completion_bonus = 0.0
if is_depot and self.state.all_serviced():
  completion_bonus = COMPLETION_BONUS

# ----------------------------
# 5) Reward components
# ----------------------------
r_drive = -float(drive) * COST_PER_DRIVE

total_weight = float(self.state.total_weight)  # lbs (after service for stops)
mpg_actual = MIN_MPG + (BASE_MPG - MIN_MPG) / (1.0 + total_weight / MPG_HALF_WEIGHT)
r_fuel = - (float(dist) / mpg_actual) * COST_PER_GAL

# Stop profit only for actual stops
r_stop = (float(s.weight) * PROFIT_PER_STOP_WEIGHT) if not is_depot else 0.0

r_total = r_drive + r_fuel + r_stop + completion_bonus
self.total_reward += r_total

return r_total
```

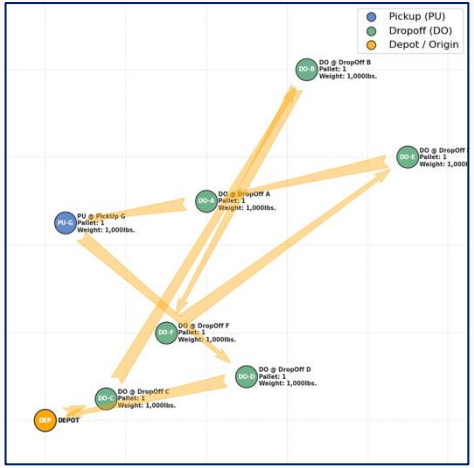
```python
# Action space:
# 0 = depot (return)
# 1..N = stops[0..N-1]
self.n_stops = len(self.stops)
self.n_actions = self.n_stops + 1
self.action_space = spaces.Discrete(self.n_actions)

# Observation space:
# rows = 1 (depot) + N (stops)
# cols:
# 0: x
# 1: y
# 2: drive_to
# 3: dist_to
# 4: weight
# 5: pallet
# 6: is_serviced (0/1)
# 7...9: type_onehot ([x,x,x])
self.obs_rows = 1 + self.n_stops
self.obs_cols = 10
self.observation_space = spaces.Box(
  low=-1e9,
  high=1e9,
  shape=(self.obs_rows, self.obs_cols),
  dtype=np.float32,
)
```

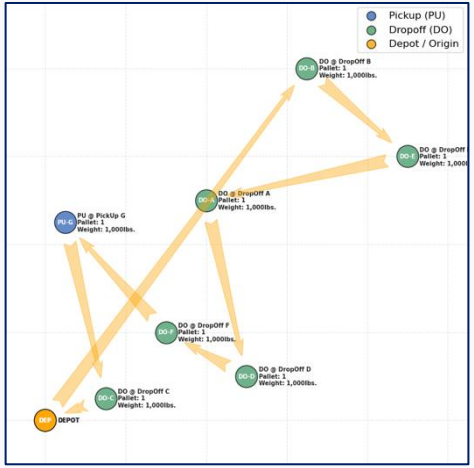
# TRAINING PROGRESS EXAMPLE

## Iterative RL Training Processes

Scenario

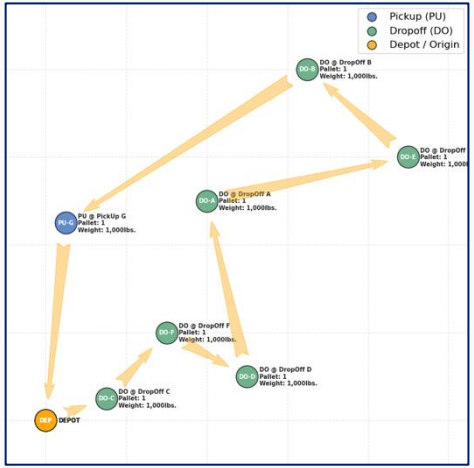
100 Episodes




250 Episodes

500 Episodes



# EARLY TAKEAWAYS FROM RESEARCH

## Lessons Learned From Applying RL to Logistics

**Reward Design Is Everything**
I noticed that the model would exploit any loophole or miss-configuration in the feedback-loop.

**Simulation Fidelity Matters**
If the environment isn't realistic; the model will not generalize to the field well.

**RL will Find Hidden Shortcuts**
During evaluation I noticed shortcuts that human heuristics have missed.

RESEARCH INSIGHT

**The quality of the model is directly tied to the realism of its training environment.**

# MAIN TAKEAWAYS RL: COMPLEX ROUTING

## The Future of Decision-Driven Operations Using RL

### RL is About Sequential Decisions

It's the perfect tool for problems where one choice affects the next.

### Reward Design Is the Bridge

It's how we translate business goals into a language the model understands.

### Practical RL Requires Realism

Success in the field depends on the fidelity of the training environment.

# QUESTIONS?

Thank you for your time!

**FEEDBACK**
Thank you!

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